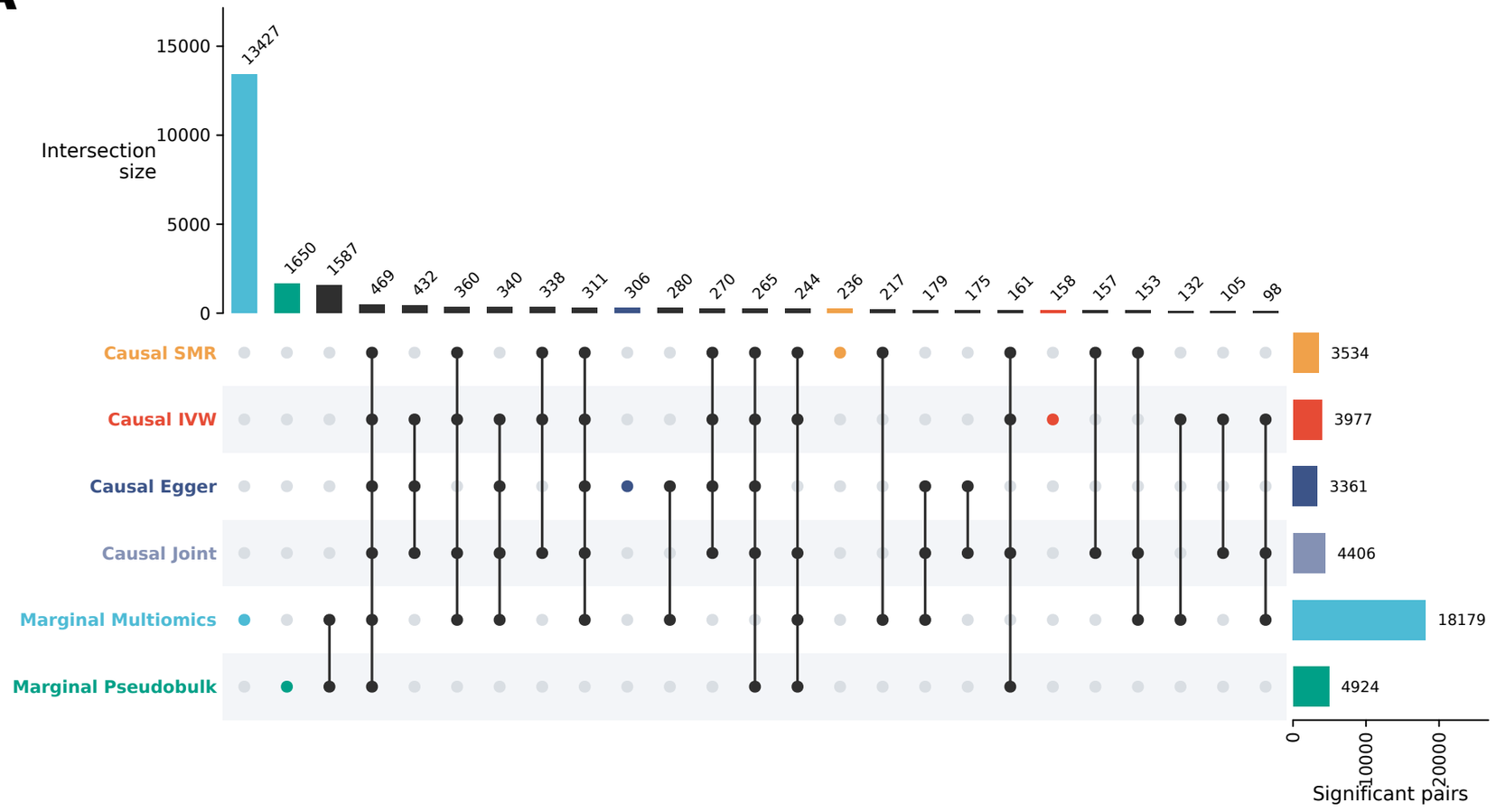
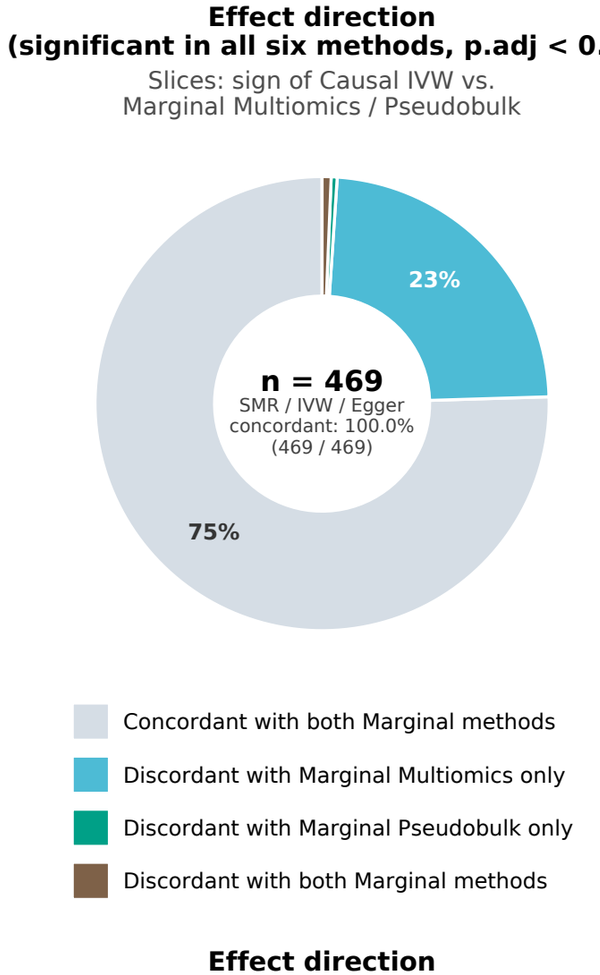
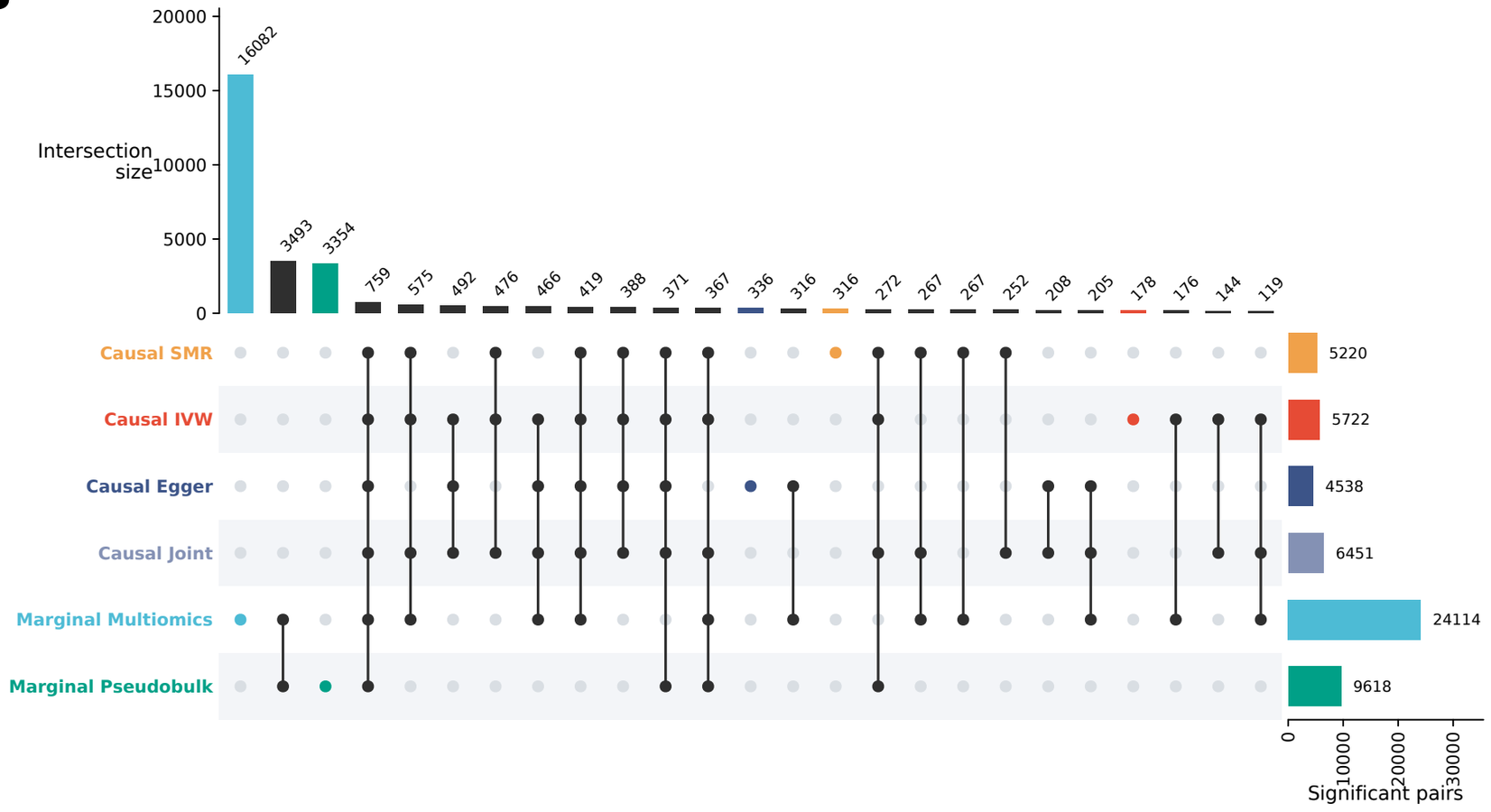
**A**


Bn_TCL1A: significant pairs across methods (p.adj < 0.1; 22,766 pairs significant in >= 1 method)
Intersection size
15000
10000
5000
0
13427
1650
1587
469
432
360
340
338
311
306
280
270
265
244
236
217
179
175
161
158
157
153
132
105
98
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
3534
3977
3361
4406
18179
4924
0
10000
20000
Significant pairs


Effect direction
(significant in all six methods, p.adj < 0.1)
Slices: sign of Causal IVW vs.
Marginal Multiomics / Pseudobulk
23%
75%
n = 469
SMR / IVW / Egger
concordant: 100.0%
(469 / 469)
Concordant with both Marginal methods
Discordant with Marginal Multiomics only
Discordant with Marginal Pseudobulk only
Discordant with both Marginal methods

**B**


CD8_Tn_CCR7: significant pairs across methods (p.adj < 0.1; 31,480 pairs significant in >= 1 method)
Intersection size
20000
15000
10000
5000
0
16082
3493
3354
759
575
492
476
466
419
388
371
367
336
316
316
272
267
267
252
208
205
178
176
144
119
Causal SMR
Causal IVW
Causal Egger
Causal Joint
Marginal Multiomics
Marginal Pseudobulk
5220
5722
4538
6451
24114
9618
0
10000
20000
30000
Significant pairs


Effect direction
(significant in all six methods, p.adj < 0.1)
Slices: sign of Causal IVW vs.
Marginal Multiomics / Pseudobulk
24%
74%
n = 759
SMR / IVW / Egger
concordant: 99.9%
(758 / 759)
Concordant with both Marginal methods
Discordant with Marginal Multiomics only
Discordant with Marginal Pseudobulk only
Discordant with both Marginal methods